



BRAC UNIVERSITY
Department of Computer Science & Engineering
FINAL EXAMINATION, FALL 2023
CSE 350: Digital Electronics and Pulse Techniques



Total Marks: 60 (20×3)

Time: **90 Minutes**

- Total no of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet
- There is no need to draw circuits in your answer sheet.

Name:	ID:	Section:
-------	-----	----------

Question 1

For the following **square wave generator** circuit, the value of C_X , R_2 , R_3 and R_4 are $0.05\mu\text{F}$, $15\text{k}\Omega$, $60\text{k}\Omega$ and $100\text{k}\Omega$ respectively. In this circuit, the **higher threshold voltage**, $V_{TH} = 5.5\text{ V}$. [20]

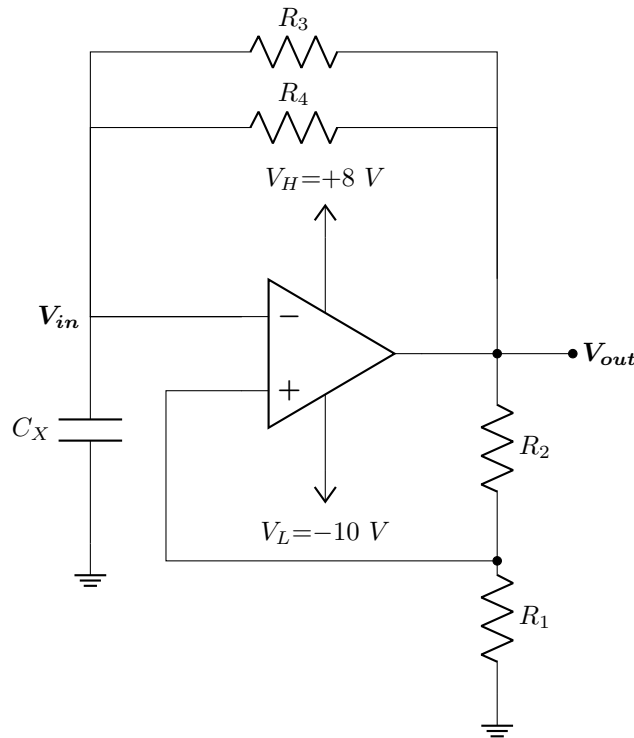


Figure 1

CO3	(a)	Find the Hysteresis Width and resistance value of R_1 .	[2.5+2.5]
CO3	(b)	Find the frequency and duty cycle of the signal generator.	[7]
CO3	(c)	Draw the graphs of (V_{in} vs t) and (V_{out} vs t) with clearly showing the maximum and minimum value of each signal and time period.	[4]
CO3	(d)	(i) If R_3 is removed, what are the effects on duty cycle and time period? (ii) If V_{TH} is increased, what are the effects on duty cycle and time period?	[2+2]

Question 2

A signal $x(t)$ is provided in **Figure 3**. A process of 'AC-DC' and 'DC-AC' conversion is done to the signal. The plot shows $x(t)$ and 5 sampled (not quantized) values. The entire process is shown in the flow chart **Figure 2**. The conversion is done using the following quantities: $V_{min} = 0$ V, $V_{max} = 2.6$ V, V_{REF} (for DAC)= 2.6 V.

CO2	(a)	Determine the Resolution, Δ . Then, find all the quantization ranges for the 'AC-DC' conversion .	[2+8]
CO2	(b)	Find the values that will be assigned to the 5 samples after quantization. Also, find the encoded version for each of them : $[b_1 \ b_2 \ b_3]$.	[2.5+2.5]
CO2	(c)	Use the encoded values of the 5 points and pass it through the DAC shown above. Do the 5 DAC outputs match with the original 5 sampled values? Show using calculations only, circuit not required.	[5]

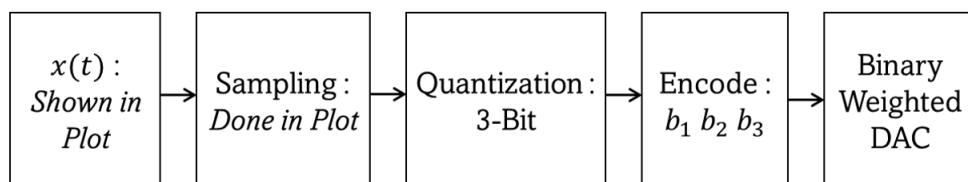


Figure 2

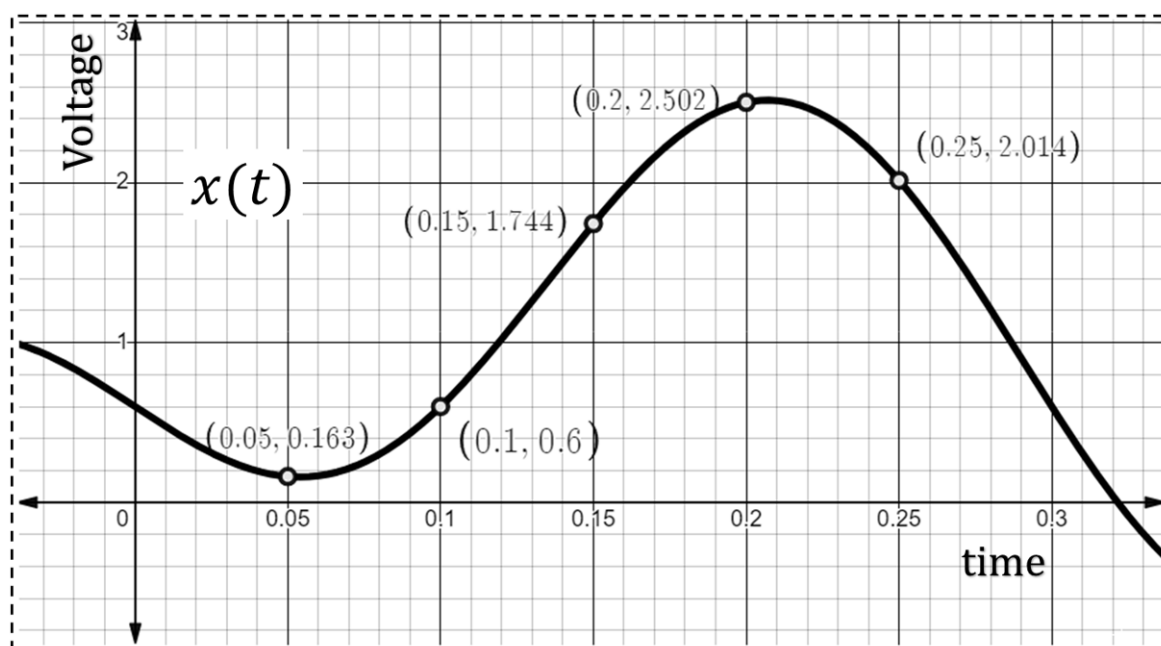


Figure 3

Question 3

Sampled values from a sinusoid are input to the dual-slope ADC converter shown in **figure 4**. The V_{REF} value is always set at the highest possible value of the analog voltage it is converting. V_{in} is the input sampled values from the analog signal. Also given:

ADC clock frequency: 2 MHz
 Counter bits (N): 4 bits
 Resistance: $R = 1 \text{ k}\Omega$
 Capacitance: $C = 1 \mu\text{F}$

The following sinusoid is sampled and given as input to the ADC: $y_1 = 2 + 2 \cdot \sin(200\pi t)$

[20]

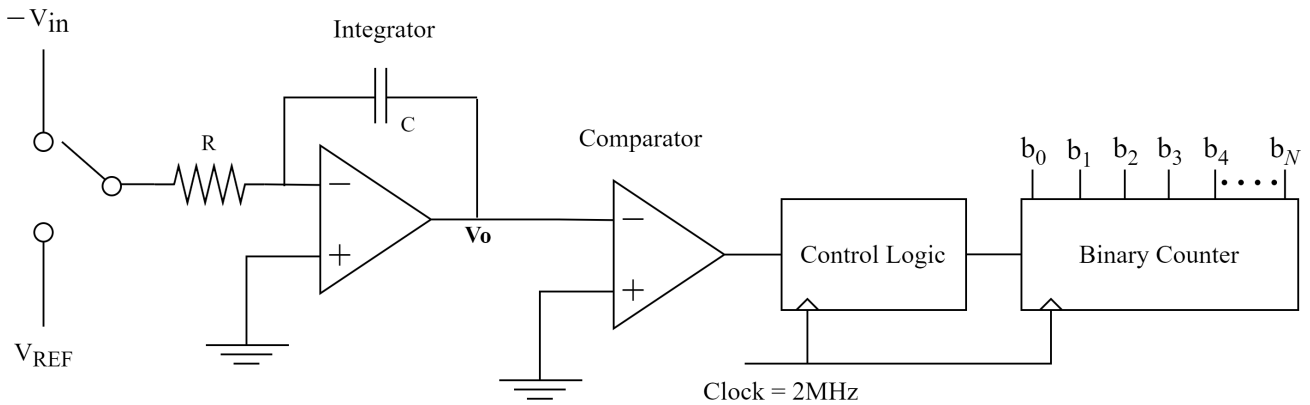


Figure 4

CO3	(a)	A sample is taken at $t = 4 \times 10^{-3}$ seconds from y_1 . Calculate the binary output value of the ADC. (Remember to keep your calculator in radian mode!)	[6]
CO3	(b)	Find the total time taken to process the value in part (b) from the ADC input to the binary output. Finally, sketch the voltage V_0 (Voltage versus Time) denoting the • slopes of the lines • Time values on the x-axis. (you do not have to label the voltage values on the y -axis)	[10]
CO3	(c)	Suppose the signal y_1 is sampled at 4000Hz. Comment with a numerical calculation whether the given ADC converter can support this sampling rate. [Hint: Calculate the maximum possible sampling rate and compare]	[4]

Question 4

The enhancement-load NMOS NOR gate in **figure 5** is biased at $V_{DD} = 5V$. The transistor parameters are $K_n = 0.3 \text{ mA/V}^2$, $V_{TN1} = 0.5V$, $V_{TN2} = 0.8V$, $V_{TNL} = 0.7V$. For triode mode, $i_D = k_n [2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2]$, and for saturation mode, $i_D = k_n [V_{GS} - V_{TN}]^2$ [20]

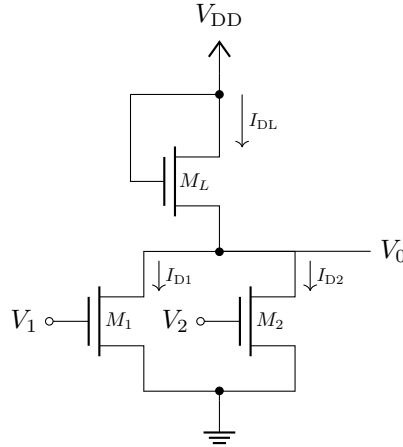


Figure 5

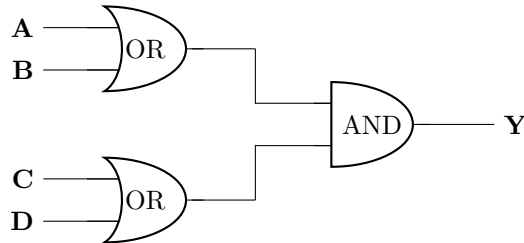


Figure 6

CO1	(a)	Find the operating mode for the logic gate of the load NMOS transistor in figure 5 .	[2]
CO1	(b)	Find the value of V_0 in V and I_{DL} , I_{D1} , I_{D2} in mA for $V_1 = 5V$, $V_2 = 5V$.	[8]
CO1	(c)	Identify the Boolean function for output, Y in figure 6 . Design the CMOS circuit for Y (Clearly show the PMOS and NMOS networks).	[2+8]